Xu Huang

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Students in school

EDUCATION

Shanghaitech University Sep 2024 - Present
Electronic Science and Technology Master Robotics center, institute of information science and technology
- Research direction: Based on deep reinforcement learning bionic robot fish motion control application

Hunan University of Science and Technology Sep 2020 - Jun 2024
Mechanical design and manufacturing and its automation Bachelor School of Mechanical and Electrical Engineering

Project and research experience

Self-developed bionic robot fish research platform Dec 2024 - Present
Project Description: in order to meet the needs of underwater complex environment detection, a 0.5m long imitation barracuda robot is independently designed and developed, and a scientific research experimental platform integrating mechanical structure, embedded hardware and control algorithm is built to realize the verification of bionic propulsion and autonomous motion ability.
- Mechanics and Modeling: Based on the multi-joint bionic structure and streamlined shape design are SolidWorks completed, and the URDF model is constructed for simulation alignment, and the static sealing technology is used to realize the IP68 waterproof of the whole machine, which supports the continuous and stable operation of the cabin system under dynamic load for more than 1 hour.
- Embedded and hardware development: based on the ESP32-S3 master to complete the hardware design and integration, design schematics and draw PCB, power distribution, communication interface and actuator control circuit. The system uses 8.4V model aircraft lithium battery power supply, combined with DC-DC voltage regulator module to build a stable power supply link. Based on CAN(TWAI), a communication network is constructed to realize unified control of multiple CAN servos and integrate IMU inertial sensors to complete state acquisition. 915MHz(CRSF) remote control link and WiFi(AsyncUDP) debugging link are designed to realize decoupling of control and debugging and multi-link parallel communication. Interface matching, communication coordination and task scheduling design are completed in the system integration, and 50Hz stable motion control is realized. The system can switch and operate reliably in laboratory and outdoor water environment.
- Motion control algorithm: Design a multi-joint collaborative propulsion algorithm based on the central pattern generator (CPG), optimize the propulsion efficiency through parameter identification, so that the cruise speed reaches 0.5 m/s, and complete the offline deployment of the algorithm. Reinforcement learning and Sim2Real verification, based on CFD fluid simulator, deep reinforcement learning control strategy training, and simulation to real machine migration method.

Inland river inspection bionic robot fish (enterprise cooperation project phase I) Sep 2025 - Present
Project description: research and develop an underwater bionic robot system with independent operation ability to meet the needs of high risk of inland ship lock inspection and difficult defect identification. Lead the design of mechanical structure and sealing scheme of the whole machine, complete the selection of power system (steering gear) and optimization of cabin structure, and complete the development of engineering prototype that can be launched from zero propulsion processing, assembly and debugging process. Participate in the development of ESP32 master control platform based on Jetson Orin Nano, complete the integration of multi-sensor system (camera, sonar, IMU, GPS, depth meter), and realize the stable acquisition and processing of multi-source perception data. Development of bionic propulsion control algorithm development, based on CPG and closed-loop PID to achieve multi-joint collaborative propulsion and depth control. Complete the prototype underwater experimental verification, with basic cruise and fixed depth capabilities, to provide platform support for the subsequent development of autonomous inspection algorithms.
- **Project results: completion of the first phase of engineering prototype development (including structural design drawings, embedded control systems and test data)**

Agile motion control of bionic robotic fish based on deep reinforcement learning Sep 2025 - Mar 2026
Project Description: Combined with CFD fluid simulator, an end-to-end intensive learning algorithm based on SAC is designed for the high mobility demand of underwater multi-joint bionic robot fish. The control delay, observation noise and random initial position and attitude are introduced to improve the generalization ability and anti-disturbance performance of the strategy. In the case of no explicit dynamic model, the rapid start-up behavior of driving robot fish to successfully reproduce the high mobility of fish in nature is realized in the simulation.
- **Project Outcome: Robotics International Conference ICRA 2026 has received one (first author)**

Deployment and Sim2Real validation of anti-flow control algorithm for bionic robotic fish Apr 2025 - Sep 2025
Project description: Based on reinforcement learning SAC robot fish flow control research. Responsible for deploying reinforcement learning control strategy to robotic fish platform and building a real-machine experimental framework based on **ROS**; 50Hz frequency synchronization of perceptual data and control instructions is realized through **UDP (WiFi)** protocol and ROS topic mechanism; **rosbag** is used to realize alignment and acquisition of multi-source data (visual pose, motion state and control amount); Participate in dynamic parameter identification and strategy optimization in Sim2Real migration process, complete the closed-loop control verification in complex water flow environment.
- **Project Results: YAC 2026 of Robot China Youth Conference has received one article, and TRO, a robot journal, is reviewing one (second author)**

Internship experience

Xiliu Digital Technology Co., Ltd. (AEROCAE) Mar 2024 - Jun 2024
Assistant Engineer (CFD Simulation and Engineering Validation)
During the internship, I participated in the CFD fluid simulation verification of the UAV wing type and automobile pneumatic model, and carried out error analysis and consistency evaluation of numerical simulation results based on experimental data, and assisted in verifying the convergence and physical rationality of the model. Daily projects include: participating in stress testing and anomaly analysis of simulation process, locating and feeding back 20 numerical calculation and logic problems.

HONORS & AWARDS

First-class scholarship at the school level (Top 5%), excellent graduation design at the school level, three good students at the school level, and one utility model patent
A total of 3 papers have been published/under review in the field of robotics: ICRA has received (one work), TRO is under review (two work) and YA C has received

SKILLS LIST

C/C , Python,ROS,Solidworks,CAD, Jialichuang EDA,3D Additive Manufacturing, ESP32,Arduino,CET-6, Driver's License C1